Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Run Exploratory Data Analysis (EDA)
2. Abbreviated Title	Run EDA
3. Use Case Id	UC-02
4. Actors	Clinician (Primary), ML Engine (Supporting)
5. Description	The Clinician initiates an EDA pipeline on the Cleveland Heart Disease dataset to visualize distributions, correlations, missing values, and feature importance before training the model.
5.1. Pre Conditions	1. The Cleveland dataset is loaded and accessible. 2. EDA dependencies (Pandas, Matplotlib, Seaborn) are installed.
5.2. Task Sequence	1. Clinician triggers the EDA module via CLI or dashboard. 2. System loads and profiles the dataset. 3. System generates statistical summaries (mean, std, missing %). 4. System renders distribution plots for each feature. 5. System renders a correlation heatmap. 6. System highlights key insights (class imbalance, outliers).
5.3. Post Conditions	1. Visualizations (histograms, heatmaps, box plots) are generated. 2. Key statistical insights are available for model training decisions.
6. Modification History	Date: September 11, 2026
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